The Cartesian product of any two sets exists

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Editorial note. The proof invokes Powersets and Separation, both axioms stated in this same chapter, and the pairs-consequences proposition of the same unit, published as corpus card X2-logic-z-singleton-binunion-pairs.

Editorial note. The author's displayed set-builder definition of C inside the proof is retained verbatim in the counted span.

Editorial scope. One original proposition of the unit's powersets section, counted once: existence of the Cartesian product of two arbitrary sets. The counted proposition and its author proof are retained in full; the follow-up discussion of Powersets interacting with Separation and the two exercises after it are not counted.

Allowed standard prerequisites (editorial).

- The Powersets axiom: for every set A the set of all subsets of A exists.
- The Separation scheme: for every set A and formula phi, the set of elements of A satisfying phi exists.
- The ordered pair construction and its existence properties from the pairs-consequences proposition of the same unit: for any sets x and y the pair {x} and {x, y} and the ordered pair (x, y) exist; published as corpus card X2-logic-z-singleton-binunion-pairs.
- The Kuratowski encoding of the ordered pair (x, y) as the set with elements {x} and {x, y}, so that (x, y) is an element of the powerset of the powerset of A union B whenever x and y lie in A union B.

Source reference. Original proposition pairsconsequences: sources/olp-set-theory-z/pinned-original-source.tex, source lines 715–715. The proof invokes the pairs-consequences proposition twice, for the existence of the ordered pair (x, y) and of its components. It is a navigation pointer to material the counted proof cites; the counted proof retains its own argument in full. The proposition is proved in the same unit and is published as corpus card X2-logic-z-singleton-binunion-pairs.

Proposition 0.1. *Given any sets A, B, their Cartesian product $A \times B$ exists.*

Proof. The set $\wp(\wp(A \cup B))$ exists by Powersets and pairsconsequences . So by Separation, this set exists:

$$C = \{\, z \in \wp(\wp(A \cup B)) \mid (\exists x \in A)(\exists y \in B) z = \langle x, y \rangle \,\}.$$

Now, for any $x \in A$ and $y \in B$, the set $\langle x, y \rangle$ exists by pairsconsequences . Moreover, since $x, y \in A \cup B$, we have that $\{x\}, \{x, y\} \in \wp(A \cup B)$, and $\langle x, y \rangle \in \wp(\wp(A \cup B))$. So $A \times B = C$. $\square$